

1. The mean SAT verbal score of next year's entering freshmen at the local college is 600. The college also knows that 69.5% of the students scored less than 625. If the scores are normally distributed, what is the standard deviation of the SAT scores?
  - a. 18.25
  - b. 49.02
  - c. 4.5
  - d. 9.5
2. When finding a confidence interval for a population mean based on a sample of size 8, which assumption is made?
  - a. The sampling distribution of  $\bar{z}$  is normal.
  - b. The sample should be taken from a normal population.
  - c. There is no special assumption made.
  - d. The population standard deviation,  $\sigma$ , is known.
3. From a sample of 200 items, 12 items are defective. The point estimate of the population proportion defective will be:
  - a. 200
  - b. 0.12
  - c. 12
  - d. 0.06
4. If all the points in a scatterplot lie on the equation of the regression line, then the correlation coefficient  $r$  must be:
  - a. either 1 or -1
  - b. 100
  - c. -1
  - d. 1
5. A random sample of size 15 taken from a normally distributed population revealed a sample mean of 75 and a sample variance of 25. The 95% confidence interval for the population mean would be closest to:
  - a. (72.23, 77.77)
  - b. (74.11, 69.88)
  - c. (71.33, 76.67)

- d. (74.33, 75.67)

6. For a certain sample, we find that

$$\sum_{i=1}^{11} (x_i - \bar{x})^2 = 90.$$

The sample standard deviation (ss) equals:

- a. 11
  - b. 9
  - c. 33
  - d. 3
7. A random sample of  $n=100$  yields a sample mean  $(\bar{x}=50)$ . Suppose that the population standard deviation  $(\sigma=20)$ . For testing  $H_0: \mu=45$  versus  $H_a: \mu > 45$ , the p-value is:
- a. 0.0062
  - b. 0.05
  - c. 0.9938
  - d. 0.5172
8. Consider a large population with a mean of 150 and a standard deviation of 27. A random sample of size 36 is taken from this population. The standard error of the sampling distribution of the sample mean is equal to:
- a. 4.17
  - b. 5.2
  - c. 4.5
  - d. 2727
9. Suppose that a t-test is being conducted at the  $\alpha=0.05$  level of significance to test  $H_0: \mu=50$  versus  $H_a: \mu > 50$ . A sample of size 20 is randomly selected. The rejection region is:
- a.  $t > 1.729$
  - b.  $t < 1.729$
  - c.  $t < -2.093$
  - d.  $t > 2.093$
10. Given the probability distribution:

**xx     0   1   2**

**f(x)f(x) aa 0.2 bb**

11. If  $E(X)=1.4$ , then the value of  $b$  is:

- ☐ a. 0.4
- ☐ b. 0.6
- ☐ c. 0.2
- ☐ d. 0.1

12. Suppose that  $P(A)=0.2$ ,  $P(B)=0.4$ , and  $P(A \cap B)=0.92$ . Then the events  $A$  and  $B$  are:

- ☐ a. Dependent
- ☐ b. Independent
- ☐ c. 0.6
- ☐ d. Disjoint

13. The dean of admissions in a large university has determined that the scores of the freshman class in a mathematics test are normally distributed with a mean of 82 and a standard deviation of 8. Based on this information, what is the probability that the mean score of a sample of 64 students is at least 80?

- ☐ a. 0.5987
- ☐ b. 0.9772
- ☐ c. 0.4772
- ☐ d. 0.0987

14. Given the following data for variables  $x$  and  $y$ :

| <b>xx</b> | <b>yy</b> |
|-----------|-----------|
| 1         | 3         |
| 2         | 7         |
| 3         | 5         |
| 4         | 11        |
| 5         | 14        |

15. The best-fitted equation line (the regression line of  $y$  on  $x$ ) is:

- a.  $\hat{y}=2.6+2x$
- b.  $\hat{y}=0.1+17x$
- c.  $\hat{y}=0.2+2.6x$
- d.  $\hat{y}=0.3+0.5x$

16. If the random variable  $X$  is binomially distributed with  $n=10$  and  $p=0.05$ , then  $P(X=2)$  is:

- a. 0.015
- b. 0.075
- c. 0.25
- d. 0.5

17. A numerical descriptive measure calculated from the sample is called:

- a. a sampling distribution
- b. a statistic
- c. a population
- d. a parameter

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Q11 | Q12 | Q13 | Q14 | Q15 |
|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|
| B  | D  | D  | A  | A  | D  | C  | C  | A  | B   | A   | B   | C   | B   | B   |